

Aditya Chunduri

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SUMMARY

Machine Learning Engineer with experience building and deploying computer vision and LLM-based systems. Strong background in PyTorch, Docker, and end-to-end ML pipelines spanning data acquisition, model training, evaluation, and inference optimization.

TECHNICAL SKILLS

- Core ML:** Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, RAG
- Languages:** Python, C++, SQL, JavaScript, Bash
- Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face
- MLOps & Systems:** Docker, Git, CI/CD, GitHub Actions, Flask, AWS
- Data & Tools:** PostgreSQL, ChromaDB, Neo4j, OpenCV, Pandas, NumPy

EXPERIENCE

- Research Assistant** Jun 2024 – Dec 2024
University of Southern California Los Angeles, California
 - Architected artery-vein classification pipelines using U-Net, attaining a 0.92 Dice score on medical benchmarks
 - Streamlined RMHAS image scraping and quality pipelines, reducing manual dataset curation time by 40%
 - Programmed modular PyTorch evaluation pipelines to accelerate experimentation on domain-specific data by 30%
- Research Intern** Jun 2021 – Jul 2021
SSN College of Engineering Chennai, Tamil Nadu
 - Conceptualized scalable Computer Vision pipelines for real-time tracking, processing 45 FPS and streamlining dataset generation
 - Fine-tuned YOLO-based CNN models to increase detection robustness by 15% mAP across varied environmental conditions
 - Coded reusable deep learning components in PyTorch and TensorFlow to standardize model development workflows

PROJECTS

- Akashic Tree** Sep 2025 – Present
Automated GenAI Media Pipeline Python, LangChain, Ollama, Stable Diffusion, Coqui TTS, MoviePy, CI/CD
 - Built a modular brief-to-video pipeline with saved artifacts per stage for traceability and batch runs
 - Added brand checks and quality gates to block off-style outputs early and reduce wasted compute
- AI Health Journal** Jan 2025 – Apr 2025
Privacy-First Mental Health Assistant Python, Flask, Ollama, LangChain, ChromaDB
 - Formulated a local RAG system using Ollama, enabling fully on-device inference with zero external API calls
 - Executed DPO fine-tuning via a 76-prompt evaluation suite to align outputs with mental health safety guidelines
 - Optimized ChromaDB indexing to ensure low-latency retrieval and real-time context injection
- MetaLearnML** May 2024 – Aug 2024
Dataset-Agnostic AutoML Engine Python, PyTorch, Scikit-learn, FastAPI, Neo4j
 - Constructed dataset-agnostic pipelines to infer task types and generate mechanized preprocessing workflows
 - Deployed a meta-learning ranker to optimize model selection across 16 PyTorch and Scikit-learn configurations
 - Tracked experiments using Neo4j and exposed performance metrics via a custom FastAPI interface

EDUCATION

- University of Southern California (USC)** Los Angeles, California
Master of Science (MS) in Applied Data Science Jan 2024 – Dec 2025
 - Coursework:* Machine Learning, Information Retrieval, Data Mining, Algorithms, Data Management, Data Engineering
- SRM Institute of Science and Technology** Chennai, Tamil Nadu
B.Tech in Computer Science and Engineering Jun 2019 – Jul 2023
 - Specialization:* Artificial Intelligence and Machine Learning

PUBLICATIONS

- Wind Power Analysis Using Machine Learning in Wind Turbines** Aug 2023
International Journal for Research in Applied Science & Engineering Technology (IJRASET)
 - Proposed a neural network framework integrating numerical weather data to predict wind energy generation